Reconceptualizing Shadow IT: Motivations, Consequences, and Governance in the Age of Shadow AI

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Zusammenfassung

Eine Kurzzusammenfassung der Vorgehensweise und der wesentlichen Ergebnisse.
Allgemeine Merkmale

- Objektivität: Es soll sich jeder persönlichen Wertung enthalten.
- Kürze: Es soll so kurz wie möglich sein.
- Verständlichkeit: Es weist eine klare, nachvollziehbare Sprache und Struktur auf.
- Vollständigkeit: Alle wesentlichen Sachverhalte sollen enthalten sein.
- Genauigkeit: Es soll genau die Inhalte und die Meinung der Originalarbeit wiedergeben.

Abstract

Obige Zusammenfassung in englischer Sprache.

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List of Abbreviations

AI Artificial Intelligence

BYOAI Bring Your Own Artificial Intelligence

BYOD Bring Your Own Device

DLP Data Loss Prevention

IT Information Technology

NIST National Institute of Standards and Technology

SAI Shadow AI

SIT Shadow IT

SLR Structured Literature Review

XAI Explainable Artificial Intelligence

1 Introduction

Dieser Teil der Arbeit sollte folgende Inhalte haben: • Einführung in die Problemstellung • Motivation und Herleitung des Themas • Aufbau der Arbeit Hinweis: Es hat sich als hilfreich erwiesen, die Einleitung mit der Zusammenfassung bzw. dem Abstract und der Schlussfolgerung zu vergleichen. Damit stellt man sicher, dass diese inhaltlich im Bezug auf Zielsetzung und Motivation übereinstimmen. Der Umfang sollte ca. 5 % der gesamten Arbeit betragen. (= ca. 2 Seiten)

2 Theoretical Foundations

In order to establish a comprehensive understanding of Shadow IT (SIT) and its further, Shadow AI (SAI), this section provides an overview of the theoretical foundations, definitions, and theoretical lenses relevant to these phenomena.

2.1 Defining Shadow IT

SIT is a multifaceted phenomenon encompassing the use, development, or deployment of Information Technology (IT) artifacts (e.g., software, hardware, or services) outside the formal control, approval, or awareness of the central IT function. For instance, Steffi Haag and Andreas Eckhardt (2017, p. 469) describe it as "hardware, software, or services built, introduced, and/or used for the job without explicit approval or even knowledge of the organization", whereas Kopper and Westner (2016, p. 1) emphasize autonomous deployment within business departments without IT involvement. Other authors similarly highlight the lack of alignment with IT governance (Käss et al., 2021, p. 4; Rakovic et al., 2020, p. 147).

Beyond these operational perspectives, SIT can be conceptualized as a structural complement to formal IT, consisting of independently developed and used systems, processes, and practices that are not known, supported, or accepted by the official IT department (Rakovic et al., 2020, p. 147). More concise definitions further highlight either the user-driven development outside central IT or the absence of approval and monitoring as defining features (Rakovic et al., 2020, p. 147; Magunduni and Chigona, 2018, p. 2).

Collectively, these definitions establish SIT as a theoretically grounded construct characterized by autonomy, decentralization, and non-sanctioned IT activity, emerging outside formal IT governance.

2.1.1 Scope

SIT encompasses a wide range of tools and solutions, including spreadsheets, databases, cloud services, mobile devices, and other ad-hoc IT solutions (Rakovic et al., 2020,

p. 145; Kopper and Westner, 2016, p. 1). These systems are typically small-scale, developed by individuals or small teams, and tailored to immediate operational needs (Rakovic et al., 2020, p. 145). Their emergence reflects both the accessibility of modern IT tools and the flexibility required by employees in dynamic work environments.

2.1.2 Conceptual Distinctions

While SIT shares similarities with related practices such as workarounds, and Bring Your Own Device (BYOD) programs, it is distinct in being unauthorized and unsanctioned. Workarounds represent behavioral adaptations to organizational constraints, whereas SIT manifests these adaptations through technological artifacts. IT consumerization, on the other hand, refers to the broader trend of user-driven technology adoption, providing a context in which SIT can emerge (Rakovic et al., 2020, p. 145; Steffi Haag and Andreas Eckhardt, 2017, p. 470).

2.1.3 Core Characteristics

SIT is not necessarily malicious or risky in intention. Users often deploy it to enhance productivity, overcome system limitations, or facilitate collaboration. Its defining features are low visibility, lack of IT support, and a tension between organizational control and individual initiative.

2.2 Shadow AI

Definition of Shadow AI

How Shadow AI differs from Shadow IT

Unique Characteristics

Examples

Emergence and Growth

Scale and Prevalence

Sociotechnical Nature and Governance Challenges

Knowledge Leakage and Data Risk

Governance Drift and Policy Gaps

Organizational Context

Attack Surface and Technical Risks

Ethical Paradox

2.3 Conceptual Background on Shadow Behavior

This section should: • briefly explain why shadow behavior exists in general • stay high-level and theory-driven • avoid listing concrete motivations

Example content: • Workarounds in organizations • Tension between control and autonomy • Informal vs. formal IT usage • Knowledge worker behavior

3 Methodology

The review follows a structured approach based on the guidelines provided by Okoli (2015, pp. 887–902) for conducting a Structured Literature Review (SLR) in the field of information systems research. The methodology is organized into four phases of two steps each: Planning, Selection, Extraction, and Execution (Okoli, 2015, p. 885).

While the original framework is designed for collaborative research settings involving multiple reviewers, this review was conducted by a single researcher. Therefore, minor adaptions were made to certain steps (e.g., drafting the research protocol, team training) to ensure methodological rigor while maintaining the integrity of the review process. The following sections describe these phases and their implementation in the context of this review.

3.1 Purpose

The first step in conducting a SLR is to define the purpose of the review, which is to identify and analyze existing literature on the phenomenon of SIT and further the development of SAI in organizational contexts. The SLR aims to synthesize insights on the motivations behind SIT and SAI, their organizational consequences, and the governance approaches employed to manage these phenomena.

3.2 Draft Protocol

The second step includes drafting a research question, a research protocol, and training the team. Since this thesis is conducted by a single person, the protocol and team training steps are not required and will therefore not be carried out further. The research questions guiding this review are as follows:

Main Research Question (RQ): How does SAI differ from traditional SIT in organizations with regard to motivational factors, consequences, and countermeasures?

To better address the research question, it will be broken down into three sub-questions and answered throughout the paper:

1. Which motivational factors known from SIT are transferable to SAI, and which additional motivation factors are specific to SAI?
2. Which organizational consequences of SAI differ from those of SIT?

3. Which countermeasures are discussed for SIT and SAI, and how do they differ in design and expected effectiveness?

3.3 Practical Screening

To make sure only relevant and high-quality literature is included in the review, practical screening criteria are defined to filter the large number of initial search results before reviewing them in detail.

For this review, only journal articles, conference proceedings, and book chapters/sections published in English between 2005 and 2026 with a focus on SIT/SAI in organizational or professional settings were included. The research design of the included studies could be empirical (quantitative, qualitative, or mixed methods), theoretical, conceptual, or review-based, as long as they provide insights relevant to the research questions. The participants in the included studies could be professionals, managers, employees in various roles (e.g., analysts, data scientists, IT staff), or end-users as long as they have a close connection to the organizational context of SIT and SAI.

3.4 Literature Search

The literature search was conducted using four search strings across four databases (AISeL, EBSCO, IEEE, and Web of Science), employing advanced search with Boolean operators (`OR` and `AND`) applied to all fields or title/abstract/keywords depending on the database. The search strings were designed to capture literature on both concepts, as well as their intersection, to ensure a broad and comprehensive coverage of relevant studies. The search strings and their purposes are summarized in Table 1. The search process was completed on March 15, 2026, and therefore includes only literature published up to that date.

Table 1: Literature Search Strings

String	Purpose	Query
S1	Shadow IT Baseline	`"shadow IT" OR "rogue IT" OR "IT workaround" OR "feral system"`
S2	Shadow AI Baseline	`"shadow AI" OR "unauthorized AI" OR "unsanctioned AI" OR "BYOAI"`
S3.1	Shadow IT × Shadow AI	`("shadow IT" OR "rogue IT" OR "IT workaround" OR "feral system") AND ("shadow AI" OR "unauthorized AI" OR "unsanctioned AI" OR "BYOAI")`
S3.2	Shadow IT × AI (broader)	`"shadow IT" AND "artificial intelligence"`

String S3.2 was included to capture literature not explicitly using the term *Shadow AI* but discussing the intersection of SIT and Artificial Intelligence (AI) technologies.

Additionally, the search was extended by backward and forward searching of the reference list of the included studies to identify any relevant literature that may have been missed in the initial database search.

3.5 Data Extraction

The remaining literature was subjected to a systematic data extraction process, where the literature was coded and sorted based on their key scope in accordance to the research questions. Based on this preliminary coding, relevant information was extracted and collected in a structured format to group similar concepts and themes together. This process was conducted iteratively, with multiple rounds of coding and extraction to ensure that all relevant information was captured and organized effectively for further analysis and synthesis in the subsequent phases of the review.

3.6 Quality Appraisal

While the practical screening criteria help filter out irrelevant studies before even reading the full-text, the review process also involves a quality appraisal to ensure the included studies meet certain standards. Therefore exclusion criteria need to be defined and applied to the extracted studies (Okoli, 2015, p. 884).

This SLR excluded studies with unsufficient theoretical contribution, unclear methodology or findings, or low prerequisites to adequately answer the research questions.

3.7 Study Synthesization

Study synthesization involves aggregating, summarizing, and interpreting the findings from the included studies to answer the research questions (Okoli, 2015, p. 899). The aggregation utilized the preliminary outline of the review, which was based on the research questions, to group similar concepts and themes, such as definitions, motivations, consequences, and countermeasures of SIT and SAI. [1]

3.8 Writing the Review

The final phase of the review process presents the results in detail, so future reviewers can understand and reproduce the process and the findings (Okoli, 2015, p. 901). Section 4 presents the results of the SLR, organized around the key themes of identified lit-

[1] To support the aggregation and summarization of relevant data, generative AI tools such as ChatGPT (OpenAI, GPT-5.3, April 2026 version) and Claude (Antrophic, Sonnet 4.5, April 2026 version) were used. All outputs were critically reviewed and revised by the author who assumes full responsibility for the final content.

erature, motivational factors, organizational consequences, and governance approaches of SIT and SAI to summarize existing literature and knowledge about these concepts.

4 Structured Literature Review Results

4.1 Overview of Identified Literature

In total, 831 records from the four databases were identified. Table 2 provides a breakdown of the search results per string and database, illustrating the initial scope of literature available for review.

Table 2: Database Search Results per String

String	AISeL	EBSCO	IEEE	WoS	Total
S1	296	227	39	94	656
S2	6	79	9	5	99
S3.1	4	2	0	0	6
S3.2	45	18	4	3	70
Total	**351**	**326**	**52**	**102**	**831**

Following the removal of duplicates (124 records) and the application of the practical screening criteria (69 records), 638 records remained for content screening. Reviewing titles and abstracts led to the exclusion of 516 records, leaving 122 records for full-text review and quality appraisal. The full-text review resulted in the exclusion of 45 studies, with 19 additionally excluded due to inaccessibility. The forward and backward searching of references added nine more studies, leading to a final set of 66 included studies for this review.

Table 3 provides a detailed overview of the whole SLR process, including the number of records included, excluded, and remaining at each stage of the screening funnel.

Figure 1 illustrates the temporal distribution of the included studies by publication year, differentiating between SIT and SAI. The results show that SIT has been an established research stream since atleast 2012, with continuous presence and fluctuating publication activity over time, peaking in 2019. In contrast, publication on SAI only begin to emerge in 2021 and remain limited in number. A noticeable increase can be observed from 2024 onwards, with a peak in 2025. Despite this recent growth, the overall volume of SAI remains substantially lower compared to SIT. Overall, the figure highlights that while SIT represents a mature and consistently researched phenomenon, SAI is a comparatively developing area of research.

Table 3: Literature Screening Process

Stage	**Included**	**Excluded**	**Remaining**
Records identified	–	–	831
After duplicate removal	–	124	707
Practical screening			
Language (English only)	–	24	683
Publication year (2005–present)	–	30	653
Publication type	–	15	638
Content screening			
Title screening	254	384	254
Abstract screening	122	132	122
Full-text review			
Not accessible	–	19	103
Full-text screening	58	45	58
Forward searching	3	–	61
Backward searching	5	–	66

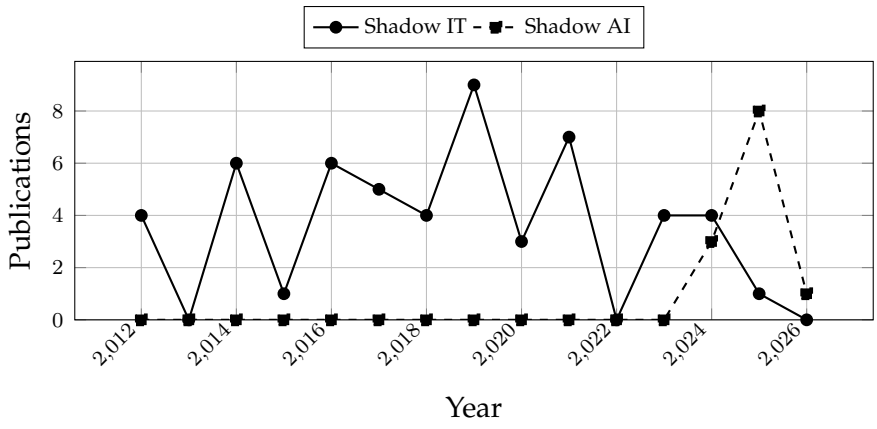

Figure 1: Temporal distribution of included studies on Shadow IT and Shadow AI (2012–2026)

4.2 Motivational Factors

The following presents the different orientations of motivational factors.

4.2.1 Shadow IT

The motivational drivers of SIT are multifaceted, spanning individual, organizational, and technological dimensions. As a systematic review establishes, there is always an underlying need for end users to complete their jobs that ultimately prompts unauthorized IT adoption (Rakovic et al., 2020, p. 148), and a comprehensive framework situates these drivers within three overarching categories: enablers, motivators, and missing barriers (Klotz, Kopper, et al., 2019, p. 22).

The most fundamental driver is the gap between what employees require and what the IT department provides. Dissatisfaction with officially sanctioned systems, combined with performance pressure, leads employees to seek unauthorized alternatives (Pinto, Beerepoot, and A. C. G. Maçada, 2023, pp. 233, 235), and SIT fills precisely this void between users and official IT solutions (Mario Silic, Barlow, and Andrea Back, 2017, p. 1023). Employees frequently perceive shadow tools as more useful than formal alternatives (G. Mallmann and A. C. Maçada, 2016, p. 3), particularly when official systems fail to align with actual job routines or adequate applications are simply unavailable (Nwankpa, 2024, p. 1). SIT also addresses entirely new functional requirements not yet covered by central IT (Chua, Storey, and Chen, 2014, p. 2), emerging as a direct result of unmet demand from functional units (Chua, Storey, and Chen, 2014, p. 11). Employees with unfulfilled technological needs actively seek and independently implement solutions (Klotz, Westner, Strahringer, and Schieder, 2019, p. 5), and the inability to fulfill business needs through legitimate channels is a primary structural driver of SIT (Györy et al., 2012, p. 6). This is compounded when business units lack the power to influence central IT priorities (Furstenau et al., 2016, p. 3), and unacknowledged grievances remain among the most cited causes of SIT across contexts (Walterbusch, Fietz, and Teuteberg, 2017, p. 653).

Workaround behavior and SIT usage positively impact individual performance by enabling faster task completion (Pinto, Beerepoot, and A. C. G. Maçada, 2023, p. 239), and represent a source of broader process innovation (White, 2023, p. 114). Employees credit shadow tools with flexibility and creativity in task accomplishment (Trialih, 2023, p. 1171) and with enabling remote data access and extended working hours (Koch et al., 2014, p. 2). Shadow systems offer the customization that rigid legacy environments deny (Myers et al., 2017, p. 108), though these short-term productivity gains can simultaneously create cybersecurity vulnerabilities (Manga and Frimpong, 2025, p. 1).

IT organization slowness is the single most prominent motivator of SIT in practitioner studies, yet remains underrepresented in academic literature (Kopper, Klotz, et al., 2019, p. 16). Excessively long response times and inadequate IT support are principal reasons employees turn to unofficial solutions (Keskin et al., 2021, p. 3), and pursuing official

channels demands strict adherence to formal development procedures and approval processes (Chua, Storey, and Chen, 2014, p. 9). From a transaction cost perspective, SIT emerges when transacting with the IT department becomes more costly than self-provisioning (Stephan Zimmermann, Christopher Rentrop, and Felden, 2014, p. 7), and employees naturally gravitate toward whichever IT alternative carries the lowest cost (Lüker, Winkler, and Kude, 2016, p. 4).

Autonomy in the workplace directly drives both workaround and SIT behavior (Pinto, Beerepoot, and A. C. G. Maçada, 2023, p. 239), and the ability to implement systems quickly and independently is the most prominent motivator in qualitative studies (Klotz, Westner, Strahringer, and Schieder, 2019, p. 5). Business units that consider themselves more capable than central IT act on this by independently procuring systems and refuse to compromise for organizational standardization (Kopper, Klotz, et al., 2019, p. 16). SIT in this form depends entirely on user functions' ability to self-provision (Chua, Storey, and Chen, 2014, p. 7), and departments adopt it fundamentally to support their own processes (Stephan Zimmermann, Christopher Rentrop, and Felden, 2014, p. 1), motivated by the organizational agility self-managed solutions provide (Kopper, Fuerstenau, et al., 2018, p. 12).

Employees who find official systems stressful adopt shadow tools as coping mechanisms (White, 2023, p. 115). When organizations fail to provide adequate resources to meet prescribed goals, structural strain arises and drives employees toward unauthorized solutions (Thatte, Grainger, and McKay, 2012, p. 5), with cognitive strain confirmed as a significant determinant of cloud-based SIT and work overload identified as an independently significant structural precursor (Ogedengbe, Talib, and Rusly, 2024, pp. 74, 76). Beyond individual pressures, social influence and social presence are significant antecedents of SIT (Gabriela Labres Mallmann, Antônio Carlos Gastaud Maçada, and Montesdioca, 2019, p. 6467), and employees are strongly influenced by peer behavior and community norms in their adoption decisions (Gabriela Labres Mallmann, Antônio Carlos Gastaud Maçada, and Andreas Eckhardt, 2018, p. 12). Supervisors who implicitly expect high efficiency may inadvertently incentivize unauthorized tool use (Gabriela Labres Mallmann, Antônio Carlos Gastaud Maçada, and Andreas Eckhardt, 2018, p. 13), and SIT spreads from individual adopters to entire workgroups following social contagion dynamics (Gabriela Labres Mallmann, Antônio Carlos Gastaud Maçada, and Andreas Eckhardt, 2018, p. 14). Employees also adopt tools that enhance social presence across activities including information exchange, decision-making, and conflict resolution (Gabriela Labres Mallmann and Antonio Carlos Gastaud Maçada, 2021, p. 428), with social drivers operating across individual, managerial, and business-level dimensions simultaneously (Godefroid, Plattfaut, and Niehaves, 2021, p. 4).

The proliferation of consumer-grade tools has increased employees' technological knowledge and lowered adoption barriers (Thatte, Grainger, and McKay, 2012, p. 8). Cloud services require no expert skills to operate (Gabriela Labres Mallmann and Antonio Carlos Gastaud Maçada, 2021, p. 436) and offer universal access to work data from any de-

vice and location (Walterbusch, Fietz, and Teuteberg, 2017, p. 644); when organizations fail to accommodate these expectations, employees fulfill them independently (Walterbusch, Fietz, and Teuteberg, 2017, p. 653). Awareness of superior consumer alternatives is itself a motivational force (Junglas et al., 2019, p. 321). A significant share of SIT behavior stems not from deliberate defiance but from unawareness of policies: carelessness, poor business-IT alignment, and organizational culture are among the most cited drivers of non-compliance, with familiarity with security policies and management influence among the strongest predictors (Silvius and Dols, 2012, pp. 2, 10). Intentional non-compliance typically occurs only when employees feel restricted or are explicitly encouraged to circumvent rules by supervisors (Silvius and Dols, 2012, p. 12). Employees frequently persist in SIT simply because they do not understand applicable policies (M. Silic and A. Back, 2014, pp. 277–278), and most do not perceive their behavior as harmful or illegal (M. Silic and A. Back, 2014, p. 279), a pattern rooted in deficient communication of IT policy (Gabriela Labres Mallmann, Antônio Carlos Gastaud Maçada, and Andreas Eckhardt, 2018, pp. 14–15).

Performance expectancy, effort expectancy, and subjective norms are the primary predictors of SIT usage intention, while personal norms and formal sanctions exert no significant influence (Nguyen, 2024, p. 500). Effort expectancy is the single strongest predictor of SIT deviance (Nguyen, 2024, p. 510): employees who believe adoption will be technically straightforward are significantly more inclined to act on the intention (Nguyen, 2024, pp. 506–507). Conversely, low self-efficacy with complex official tools reduces perceived policy value and lowers resistance to unauthorized alternatives (Welck et al., 2017, p. 4). Organizational and individual inertia perpetuate SIT by embedding behavioral habits that resist redirection (Mario Silic, 2019, p. 114). At the institutional level, limited IT knowledge within business units, poor business-IT communication, and cost pressures within departments all create conditions conducive to SIT (Furstenau et al., 2016, pp. 5–7), while workarounds arise from obstacles to task completion or goal misalignment between stakeholders (Roeder et al., 2014, p. 2). SIT also represents active resistance to mandated systems (Magunduni and Chigona, 2018, p. 3), implemented to address individual or group needs rather than those of the organization as a whole (Magunduni and Chigona, 2018, p. 5). Finally, remote work amplifies these motivations: the absence of peer monitoring reduces perceived compliance norms, remote settings provide a convenient external attribution for rule-breaking, and employees who lack guilt over deviant behavior face lower social inhibition against non-compliance (N and Mathew, 2023, pp. 4, 5, 7).

4.2.2 Shadow AI

The motivational drivers of SAI bear a structural resemblance to those documented for SIT, yet are shaped by the distinctive characteristics of AI technology itself. At the most fundamental level, employees are motivated by a pressing demand for productivity gains, using unsanctioned AI tools to significantly enhance output, automate routine tasks, and augment decision-making (Ross, Hibbert, and Moss, 2025, pp. 2, 3). Genera-

tive AI in particular is adopted for concrete time-saving purposes, drafting emails summarizing documents, and brainstorming ideas, that official tools do not readily support (Ross, Hibbert, and Moss, 2025, p. 4). This efficiency orientation is consistently confirmed across studies: AI is widely perceived as beneficial for automating repetitive tasks and providing analytical insights (Mario Silic, D. Silic, and Kind-Trüller, 2025, p. 7), with interviewees describing AI-powered tools as transformative for workflow efficiency (Mario Silic, D. Silic, and Kind-Trüller, 2025, p. 9), and employees in metaverse-based business contexts claiming that SAI enables them to be more efficient, productive, and innovative (Chin et al., 2025, p. 2). Productivity gains are also a key driver in sensitive industries: employees in healthcare, finance, and education seek out third-party AI tools to enhance performance (Balogun et al., 2025, p. 2), driven by the promise of greater efficiency and agility across those sectors (Balogun et al., 2025, p. 16). The broader productivity proposition is further reinforced by AI tools' promise of greater efficiency and innovation and new opportunities for competitive advantage (Puthal, Ray, and Mohanty, 2025, p. 973).

A second major motivational cluster mirrors the unmet needs dynamic established in the SIT literature. SAI emerges as a direct response to perceived inefficiencies in officially sanctioned digital infrastructures (Ross, Hibbert, and Moss, 2025, p. 2), driven by unmet needs, bureaucratic inertia, and a lack of accessible sanctioned alternatives (Ross, Hibbert, and Moss, 2025, p. 6). Like SIT before it, SAI is fueled by bureaucratic bottlenecks, centralized system unresponsiveness, and misalignment between official IT infrastructure and actual operational needs (Ross, Hibbert, and Moss, 2025, p. 1). Its informal adoption is frequently an operational necessity arising from limited technical resources (Ross, Hibbert, and Moss, 2025, p. 1), and governance frameworks often lag behind employee practices, creating a governance drift zone where formal policies exist but carry no real-world traction (Mario Silic, D. Silic, and Kind-Trüller, 2025, p. 1). Many workers feel that the agility gained through AI experimentation justifies bypassing slower official approval processes, and employees increasingly bypass traditional IT procurement altogether to adopt sophisticated LLMs for daily tasks (Raizada, Shakya, and Rahul, 2026, p. 1136). SAI emerges in practice where employees circumvent formal IT protocols for personal productivity gains, a dynamic not fully captured by existing adoption frameworks (Mario Silic, D. Silic, and Kind-Trüller, 2025, p. 3). Shadow systems recurrently emerge due to policy vacuums, IT bottlenecks, or employee frustration (Raizada, Shakya, and Rahul, 2026, p. 1138), and vague approval mechanisms and low policy awareness further facilitate risk-taking behavior (Raizada, Shakya, and Rahul, 2026, p. 1140). Employees who lack formal guidance or perceive policies as absent will frequently conceal their AI usage (Varshini and Sampathkumari, 2025, p. 2).

Behavioral and cultural factors form a third distinct motivational dimension. SAI usage is predominantly efficiency-driven rather than malicious, especially under task pressure (Raizada, Shakya, and Rahul, 2026, p. 1140), and the psychological factors of perceived productivity and psychological safety frequently outweigh compliance concerns for the average employee (Raizada, Shakya, and Rahul, 2026, p. 1136). Employees who expe-

rience task pressure report significantly different perceptions of workload when using unsanctioned AI tools (Varshini and Sampathkumari, 2025, p. 5), and AI has been credited with sparking creativity and enabling new approaches to problem-solving (Mario Silic, D. Silic, and Kind-Trüller, 2025, p. 9). Employees also turn to unsanctioned AI to fill gaps in official resources and innovate beyond existing constraints (Varshini and Sampathkumari, 2025, p. 1). Adoption is further fueled by a culture of digital improvisation in which employees feel compelled to find their own solutions (Ross, Hibbert, and Moss, 2025, p. 1), shaped by cultural desires for innovation, resistance to bureaucracy, and the informal networks through which digital tools are shared (Ross, Hibbert, and Moss, 2025, p. 3). This dynamic creates a culture where innovation effectively bypasses control, making top-down prohibitions largely ineffective and driving usage further underground (Raizada, Shakya, and Rahul, 2026, p. 1136), with innovation-seeking and psychological safety identified as the key behavioral drivers of AI experimentation with unmonitored tools (Raizada, Shakya, and Rahul, 2026, p. 1138).

Accessibility and skill gap considerations round out the motivational picture. SAI tools are not restricted to data science specialists, marketing, human resources, and engineering employees adopt them equally, and the availability of ready-made models in repositories enables non-experts to rapidly prototype and deploy sophisticated models without technical expertise (Puthal, Ray, and Mohanty, 2025, p. 973). Employees also resort to consumer-grade AI solutions to compensate for skill gaps within their teams (Ross, Hibbert, and Moss, 2025, pp. 1–2) and use AI skills to offset specific technical weaknesses (An, Lin, and Luo, 2024, p. 3). In the era of the digital economy, rapid technological innovation and globalization make AI adoption a competitive necessity for organizational survival (An, Lin, and Luo, 2024, p. 1), and the broad range of tasks to which digital agents can be applied, from decision-making and communication to information processing and value creation (Alter, 2024, p. 11), further explains why employees perceive AI adoption as an individual imperative even absent organizational sanctioning. The Bring Your Own Artificial Intelligence (BYOAI) context moreover links user motivations directly to governance design, as a structured mapping of motivations to governance responses reveals clear patterns in how employee needs drive both adoption and the demand for legitimate alternatives (Anthuvan et al., 2025, pp. 2708, 2709).

4.3 Organizational Consequences

4.3.1 Shadow IT

SIT consequences are fundamentally dualistic, generating both significant organizational risks and genuine performance benefits. Research documents outcomes ranging from security breaches and governance failures to productivity gains and innovation, and SIT instances can function either as replacements for official solutions or as complementary additions, depending on context (C. Rentrop and S. Zimmermann, 2012, p. 1025). IT managers must therefore balance these competing consequences when formulating governance responses (Gabriela Labres Mallmann, Antônio Carlos Gastaud

Maçada, and Andreas Eckhardt, 2018, p. 13).

SIT undermines official systems and can damage organizational data and processes (Mario Silic, Barlow, and Andrea Back, 2017, p. 1025), while unauthorized workarounds compromise information security more broadly (White, 2023, p. 114). Three structural mechanisms generate exposure: shadow systems fall outside IT security oversight, they expand the organizational attack surface, and sensitive data stored or transmitted through unsecured tools creates material breach risk (Nwankpa, 2024, p. 1). Cloud shadow tools fall outside IT maintenance cycles, leaving security vulnerabilities from malware and spyware undetected (Walterbusch, Fietz, and Teuteberg, 2017, p. 653), and employees who bypass formal systems are more susceptible to phishing attacks (Manga and Frimpong, 2025, p. 1). SIT is consequently one of the most significant concerns in organizational information security, with employees' unawareness of potential harm compounding the problem (Nguyen, 2024, p. 500), and practitioners identify data security as its most significant risk (Kopper, Klotz, et al., 2019, p. 10). Unauthorized cloud use creates monetary and reputational damage risk (Walterbusch, Fietz, and Teuteberg, 2017, pp. 644–645) and potential GDPR violations, data loss, and exposure to cyberattacks when sensitive data is processed on third-party servers (Steffi Haag and Andreas Eckhardt, 2024, p. 400). Unapproved software can introduce malware, and information leakage represents a pervasive privacy risk (M. Silic and A. Back, 2014, p. 280), with data integrity and information leakage recognized as the highest IT risks associated with SIT (M. Silic and A. Back, 2014, p. 278). Unauthorized device use also elevates intellectual property loss risk (Pirani and Meister, 2014, p. 3), and non-compliance carries direct legal consequences including fines and personal liability for senior management (Silvius and Dols, 2012, p. 1). User non-compliance with IT policy is widely considered the single largest information security threat facing organizations (Györy et al., 2012, p. 6).

SIT results in a loss of managerial control, reduced productivity, and deviations from intended business processes (Roeder et al., 2014, p. 1), threatening the performance improvements organizations expect from official IS investments and creating hazards and errors in critical workflows (Roeder et al., 2014, pp. 1, 3; Shaikh, 2021, p. 4). The cumulative effect is an elevated organizational risk profile and expanding technical debt (White, 2023, p. 115). At the organizational level, SIT generates inefficiencies through lost synergies, fragmented IT landscapes, data inconsistencies, and integration problems (Kopper, Fuerstenau, et al., 2018, pp. 3, 8). A comprehensive catalogue of inadequately managed SIT consequences includes time loss, rework, unconsolidated investments, data losses, infrastructure incompatibilities, audit findings, standardization deficits, erosion of trust in IT systems, and service interruptions (Ozkan et al., 2021, p. 283). Indirect costs include potentially unrecoverable organizational data loss (Ogedengbe, Talib, and Rusly, 2024, p. 75).

As shadow systems grow, IT decentralization frequently follows as a direct organizational consequence, shifting control from central IT to business units (Furstenau et al., 2016, p. 6). IT departments cannot feasibly monitor every device and application em-

ployees wish to adopt (Koch et al., 2014, p. 3), and BYOD and workaround practices force IT departments to delegate management responsibility to users while still bearing responsibilitycfor environmental security (Dulipovici and Vieru, 2016, p. 2). Shadow systems often grow beyond the capacity of the originating department to maintain, yet remain undetected for extended periods, and are frequently only discovered when multiple departments have independently developed similar solutions (Koch et al., 2014, pp. 9–10). Governance responsibility evolves dynamically over the lifecycle of a shadow instance (Klotz, Westner, and Strahringer, 2020, p. 10), and the reverse transition from Business-managed IT back to SIT is rare, as it would require stakeholders to progressively lose awareness of the system (Klotz, Westner, and Strahringer, 2020, p. 12). Paradoxically, strict prohibition may push SIT deeper into organizational secrecy rather than reducing it (Steffi Haag and Andreas Eckhardt, 2017, p. 472).

SIT also generates genuine positive consequences. It can function as an efficient substitute for formal systems and mediate the relationship between workarounds and improved individual performance (Pinto, Beerepoot, and A. C. G. Maçada, 2023, p. 242; Mario Silic, Barlow, and Andrea Back, 2017, p. 1025), providing positive value precisely where official IT constraints impede job performance (S. Haag, A. Eckhardt, and Schwarz, 2019, p. 739). SIT can boost organizational efficiency and effectiveness (Trialih, 2023, p. 1171), and empirical cases document measurable productivity improvements and explicit recognition of innovative value by IT managers (Kopper, Fuerstenau, et al., 2018, p. 8). Business-managed IT is associated with productivity gains, innovation, and improvements in agility and cross-unit collaboration (Godefroid, Plattfaut, and Niehaves, 2021, p. 4), and employee feedback arising from unmet needs can serve as a productive signal for successive improvement of official systems (Walterbusch, Fietz, and Teuteberg, 2017, p. 653). Shadow systems also enable faster, more effective communication and collaboration with employees, clients, and partners (Gabriela L. Mallmann, Antonio Carlos Gastaud Maçada, and Oliveira, 2018, p. 26), and empirical evidence confirms that social presence mediates the relationship between SIT usage and individual performance, with SIT having a significant positive relationship with perceived sensitivity and comprehension in collaboration (Gabriela Labres Mallmann and Antonio Carlos Gastaud Maçada, 2021, p. 427). The social dimension also explains how SIT facilitates broader knowledge dissemination across organizational boundaries (Gabriela Labres Mallmann, Antônio Carlos Gastaud Maçada, and Montesdioca, 2019, p. 6468), and shadow solutions additionally support knowledge sharing by enabling faster information exchange with internal and external partners (Klotz, Westner, Strahringer, and Schieder, 2019, p. 2). Radical innovations may need to deliberately violate organizational standards in order to succeed (Roeder et al., 2014, p. 1), and shadow systems that address operational gaps can evolve into parallel IT infrastructures developed faster and more practically than official processes allow (Keskin et al., 2021, p. 3). BYOD-enabled workarounds can likewise produce demonstrable business process improvements first at the individual and then at the organizational level (Dulipovici and Vieru, 2016, p. 2), and flexible IT strategies such as BYOD can enhance individual IT innovation

capacity irrespective of uptake (Köffer et al., 2015, p. 373).

Reports produced by shadow systems are consistently perceived as less credible than those from official systems, with direct implications for managerial decision quality (Myers et al., 2017, p. 106). At the individual level, employees who use SIT receive significantly lower bonus assessments than those using company-endorsed tools, even when outcomes are equivalent, and SIT use is likely to be viewed negatively in subjective performance evaluations regardless of work quality (Allen et al., 2019, p. 30). Complementarily, policies accommodating IT consumerization are associated with higher policy understanding but lower satisfaction, and compliance behavior is shaped by both procedural justice and rational cost-benefit calculations, reflecting the fundamental tension between organizational control and individual autonomy that defines the SIT consequence landscape (Lüker, Winkler, and Kude, 2016, p. 1).

4.3.2 Shadow AI

The consequences of SAI span cybersecurity vulnerabilities, regulatory and ethical failures, governance erosion, and organizational knowledge risks, while also encompassing more nuanced impacts on workplace trust and culture. The proliferation of SAI systems poses significant cybersecurity risks, including data breaches, compliance violations, and operational disruptions (Whitford, Marcel, and McCall, 2025, p. 1), with key risk domains including data leakage, model hallucination, and compliance breaches identified empirically (Raizada, Shakya, and Rahul, 2026, p. 1136). At the technical level, shadow backdoor models can be engineered to facilitate adversarial attacks, while shadow data leak models can channel sensitive business information to unauthorized recipients (Puthal, Ray, and Mohanty, 2025, p. 973). SAI introduces systemic vulnerabilities that challenge traditional cybersecurity models (Ross, Hibbert, and Moss, 2025, p. 1) and expands the attack surface for cybercriminals, making it a critical concern for security professionals (Balogun et al., 2025, p. 5). Unauthorized AI usage also raises data sovereignty concerns, as sensitive information is likely to be moved to and stored in locations that do not meet regulatory standards (Puthal, Ray, and Mohanty, 2025, p. 2).

The regulatory and ethical consequences of SAI are particularly acute in high-stakes sectors. In finance, unregulated AI-driven credit scoring systems have produced discriminatory lending practices and attracted regulatory scrutiny, and the unchecked expansion of SAI risks entrenching systemic biases that further marginalize vulnerable populations (Balogun et al., 2025, p. 3). In healthcare, unregulated AI diagnostic tools risk patient confidentiality breaches, misdiagnoses, and legal liability for institutions, while in education, unauthorized AI contributes to academic dishonesty, flawed grading, and violations of student privacy (Balogun et al., 2025, p. 2). SAI also exposes responsibility gaps in high-risk fields: the black-box nature of many AI systems makes it difficult to determine how decisions are made, undermining accountability (Mario Silic, D. Silic, and Kind-Trüller, 2025, p. 1; Balogun et al., 2025, p. 4). At the organizational

level, SAI leads to a governance drift zone in which formal policies lack real-world enforcement traction (Mario Silic, D. Silic, and Kind-Trüller, 2025, p. 1), and the deployment of BYOAI creates structural tensions between innovation, trust, and regulatory oversight (Anthuvan et al., 2025, p. 2693).

Beyond technical and regulatory exposure, SAI carries significant organizational and social consequences. Employees' use of SAI not only generates operational benefits but also poses risks of organizational knowledge leakage (Chin et al., 2025, p. 1), as sensitive organizational data processed through unsanctioned tools risks exposure outside the organizational boundary. SAI also affects workplace culture and trust, with repercussions for team dynamics and the perceived trustworthiness of colleagues (Varshini and Sampathkumari, 2025, p. 1). These social and cultural consequences extend the impact of SAI beyond the IT security domain into the relational fabric of organizational life, underscoring that governance responses must address not only technical risks but also the human and cultural dimensions of unauthorized AI adoption.

4.4 Governance Approaches and Countermeasures

4.4.1 Shadow IT

Governing SIT requires IT managers to first understand its mechanisms, causes, and consequences before any effective response can be formulated (Steffi Haag and Andreas Eckhardt, 2017, p. 469; Gabriela Labres Mallmann, Antônio Carlos Gastaud Maçada, and Montesdioca, 2019, p. 6467). The literature consistently establishes that no single countermeasure is sufficient and that effective governance requires a portfolio of complementary approaches spanning deterrence, governance frameworks, technical controls, formalization strategies, and cultural change.

A foundational finding is that purely restrictive approaches are largely ineffective. Research shows no statistically significant difference between policies that prohibit consumer device use and a complete absence of policy, leading to the recommendation that organizations implement policies that explicitly permit rather than prohibit private device use (Klesel et al., 2019, p. 856). Organizations cannot rely solely on regulations to neutralize SIT, and sanctions risk being counterproductive when employees adopt SIT in good faith to complete their work (Trialih, 2023, p. 1171): a pure punishment focus has been shown to be ineffective in hindering violation behavior (Nguyen, 2024, p. 511). Prohibition through information security policies is likely to produce abuse rather than compliance (Welck et al., 2017, p. 2). The technical extreme, eliminating discretionary budgets, locking down devices, and blocking internet access, is neither feasible nor acceptable in most modern organizations (Chua, Storey, and Chen, 2014, p. 11). Since SIT represents a form of decentralized computing outside IT service management (Gabriela Labres Mallmann and Antonio Carlos Gastaud Maçada, 2021, p. 428), IT leaders must instead accept that SIT cannot be entirely controlled or eliminated as new digital technologies emerge (Steffi Haag and Andreas Eckhardt, 2024, p. 408).

Despite these limitations, sanctions form one legitimate component of a broader governance strategy. Formal sanctions, explicit penalties such as salary reductions or negative performance reviews, provide direct deterrence, complemented by informal sanctions in the form of social peer disapproval for norm violations and the deterrent effect of shame when wrongdoings become known to others (Mario Silic, Barlow, and Andrea Back, 2017, pp. 1024, 1026, 1027). In documented cases, users were notified of non-compliant actions via email, unauthorized software was removed by IT, and users were reported to line managers (Mario Silic, Barlow, and Andrea Back, 2017, p. 1029). Sanctions are most effective when paired with rewards for compliance, with the advantages of conformity actively communicated and the costs of non-compliance, including monetary penalties or dismissal, made explicit (Roeder et al., 2014, p. 11; S. Haag, A. Eckhardt, and Schwarz, 2019, p. 739). Social pressure generated when unauthorized behavior is perceived as undesirable significantly reduces the likelihood of SIT being triggered, and an ethical organizational climate strengthens perceptions of both formal and informal controls while facilitating security policy adherence (Nguyen, 2024, pp. 506, 500, 510).

Most organizations deploy formal risk management tools, IT service management, IT governance, and security management, to direct and control business unit IT activity, with documented responses including formal compliance policies, the appointment of process owners, IT budget consolidation, and deliberate reduction of business unit IT spending (Furstenau et al., 2016, pp. 1, 7). At the portfolio level, governance distinguishes between general governance applying to all SIT and instance governance for identified Business-managed IT systems, with instance governance allocatable to the IT organization, shared between IT and the business unit, or retained within the business unit (Klotz, Kopper, et al., 2019, pp. 17, 30). Once a shadow instance is identified, organizations must decide whether to decommission or continue it, and if continued, assign governance to IT, the business unit, or a shared model (Kopper, Klotz, et al., 2019, p. 3). Three strategic governance postures have been identified: the IT-control approach, which centralizes all IT budgets and compliance; the user-driven approach, which redistributes IT innovation responsibility to business users; and the user-oriented approach, which synthesizes both while maintaining transparency and permitting a healthy extent of SIT as an innovation enabler (Györy et al., 2012, pp. 8–9). Managers must ultimately choose between near-total repeal of restrictive policies or rigorous compliance enforcement (S. Haag, A. Eckhardt, and Schwarz, 2019, p. 739). A co-governance model, allocating task responsibilities between IT and the business unit based on risk and transaction cost dimensions, has been proposed as a resolution, as controlled SIT simultaneously reduces risk and increases efficiency (Stephan Zimmermann, Christopher Rentrop, and Felden, 2016, p. 9). Organizations should also divide IS into subtasks and allocate responsibilities accordingly, transferring hardware and database components to IT while preserving business creativity (Kopper, Fuerstenau, et al., 2018, pp. 5, 8).

Practical technical controls range from simple web and network traffic monitoring to strict domain policies preventing software installation without IT authorization (M. Silic and A. Back, 2014, p. 278). At the technical frontier, machine learning applied

to time-aggregated network traffic features has demonstrated promising NAT detection capability, with time window duration shown to significantly affect detection accuracy (Nassar et al., 2021, p. 6). The establishment of a company-wide cloud security strategy, combined with an anonymous employee survey to map current exposure, is recommended as a foundational governance step (Walterbusch, Fietz, and Teuteberg, 2017, p. 644). Monitoring and identification are strategically important because rendering shadow instances overt transitions them from uncontrolled systems into governed Business-managed IT (Klotz, Kopper, et al., 2019, pp. 29–30). Early IT departments also employed direct technical controls such as hardware and software standardization and active removal of unauthorized applications from networks, though such measures are insufficient as standalone strategies (Koch et al., 2014, p. 9).

Among the most constructive countermeasures is the active formalization of identified shadow systems. Organizations must officially recognize, accept, and transfer shadow instances to a controlled area, channeling the energy driving SIT into formally supported Business IT that improves processes (Keskin et al., 2021, p. 4). When effective organizational rules are combined with education, training, awareness, and monitoring, employees transition from unauthorized to authorized use; absent these complements, unauthorized behavior persists, and the trigger for transition differs depending on the employee's current behavioral state (Pirani and Meister, 2014, pp. 6–7, 5). Assessment of a shadow system's innovative potential and its relationship to existing enterprise infrastructure should inform formalization decisions, with risk-critical instances prioritized for integration with official systems (Rakovic et al., 2020, pp. 150, 151). Although BYOD policies and ratification of unauthorized technologies represent attempts to govern SIT through formalization, they are acknowledged to be more complex in practice than in principle (Gabriela Labres Mallmann and Antonio Carlos Gastaud Maçada, 2021, p. 427).

Organizations must continuously improve employee awareness of IT security risks and applicable policies (Silvius and Dols, 2012, p. 12), as non-compliance risk is materially mitigated by training, clearly communicated policies, and regular security audits (Györy et al., 2012, p. 6). Stricter controls combined with better education and simpler policies would make SIT risk mitigation significantly more successful (M. Silic and A. Back, 2014, p. 279), and cloud-specific risks must be addressed through dedicated training programs (Walterbusch, Fietz, and Teuteberg, 2017, p. 654). Foundational cybersecurity compliance practices include hiring employees with a security-positive orientation, demonstrating leadership commitment, building practical skills through training, supporting compliance follow-through, and communicating the benefits of compliance proactively (Cram, Proudfoot, and D'Arcy, 2020, p. 195). Policy compliance is more reliably achieved when the development process and content of policies are made transparent through active communication (Lüker, Winkler, and Kude, 2016, p. 3), and IT managers must create initiatives that actively engage users in security policies (Gabriela L. Mallmann, Antonio Carlos Gastaud Maçada, and Oliveira, 2018, p. 26).

The IT department shares responsibility for SIT by failing to provide appropriate systems for employees (Gabriela L. Mallmann, Antonio Carlos Gastaud Maçada, and Oliveira, 2018, p. 26), and understanding the antecedents of SIT offers IT managers an opportunity to proactively provide suitable tools (Gabriela Labres Mallmann, Antônio Carlos Gastaud Maçada, and Andreas Eckhardt, 2018, p. 14). Acting fast and anticipating user needs is one of the most effective practitioner strategies for reducing SIT adoption (Kopper, Klotz, et al., 2019, p. 14), and the continuous monitoring of user needs to ensure suitable solutions are provided is a primary management recommendation (Gabriela L. Mallmann, Antonio Carlos Gastaud Maçada, and Oliveira, 2018, p. 26). IS design principles should be established to reduce the workarounds that official systems generate (Roeder et al., 2014, p. 3), and organizations should create a culture in which employees can openly surface their workarounds with managers, with appropriate communication channels and forums where benefits and risks can be assessed without prejudice (White, 2023, p. 120). Managers who understand the consequences of neutralizing SIT can design more effective and efficient governance strategies (Trialih, 2023, p. 1172), while a comprehensive practitioner framework recommends accepting SIT, assessing its threat to business continuity, enforcing cybersecurity policies with care, ensuring employees do not feel blocked, creating user experience teams, and even rewarding beneficial SIT usage (Steffi Haag and Andreas Eckhardt, 2024, p. 409). Managers should also assess why SIT is used in their segments and what existing controls maintain the credibility of information it produces (Myers et al., 2017, p. 106). Since IT tools evolve continuously, organizations are compelled to re-evaluate their infrastructure regularly to ensure ongoing alignment with business needs (Köffer et al., 2015, p. 373). Organizations should also increase agility and work to shift individual perceptions of the costs of SIT systems, with pilot studies providing supervised SIT access offering a means to build evidence-based governance understanding (Mario Silic, 2019, p. 116). Clear policies governing cloud IT interaction are needed as organizational structural factors and cognitive strain are independent drivers of adoption (Ogedengbe, Talib, and Rusly, 2024, p. 74). Managers deliberating governance must weigh the risk that abandoning SIT constrains employee creativity and efficiency, with the persistence of negative attitudes toward SIT even in the face of positive outcomes suggesting that institutional resistance may be more deeply entrenched than anticipated (Allen et al., 2019, pp. 31, 38).

4.4.2 Shadow AI

As with SIT, the governance of SAI demands a multi-layered response that spans technical controls, governance frameworks, regulatory compliance, and cultural change. On the technical side, automated AI discovery, inventory management, and advanced governance policy are identified as strategic countermeasures that establish organizational visibility over shadow systems (Puthal, Ray, and Mohanty, 2025, p. 973). The most advanced technical approach is Zero Trust Architecture, which treats all AI assets as potential threats and requires constant validation and access control for each asset (Puthal, Ray, and Mohanty, 2025, p. 1). Effective risk management must also integrate

technological safeguards such as AI-aware Data Loss Prevention (DLP), content filtering, data protection mechanisms, and AI traffic monitoring (Puthal, Ray, and Mohanty, 2025, p. 2), while automated monitoring, anomaly detection, and policy enforcement mechanisms have been evaluated as effective tools for governing SAI at scale (Whitford, Marcel, and McCall, 2025, p. 1). Transparency within AI systems is additionally critical: Explainable Artificial Intelligence (XAI) enhances transparency by making AI decision-making interpretable, enabling organizations to identify and rectify biases or errors produced by shadow systems (Balogun et al., 2025, p. 7).

At the governance framework level, organizations must establish comprehensive AI governance structures incorporating compliance audits, AI ethics boards, and regulatory enforcement mechanisms (Balogun et al., 2025, p. 3). A typology of SAI use cases combined with a risk classification framework has been proposed to enable organizations to categorize and prioritize governance responses (Ross, Hibbert, and Moss, 2025, p. 1). The BYOAI-GOVT framework offers a behavior-aware governance model that integrates task-risk zoning, distinguishing between Enable, Regulate, and Restrict categories, with a four-level organizational maturity scale, providing a structured path from informal adoption to responsible innovation (Anthuvan et al., 2025, p. 2693). Regular AI compliance audits and risk assessments, as recommended by the National Institute of Standards and Technology (NIST), help organizations systematically identify vulnerabilities in their AI landscape (Balogun et al., 2025, p. 7), and regulatory agencies are urged to strengthen oversight by imposing stricter penalties for unauthorized AI usage (Balogun et al., 2025, p. 3). At the supranational level, the European Union's Artificial Intelligence Act categorizes AI applications by risk level and imposes compliance obligations that strengthen organizational governance (Balogun et al., 2025, p. 7).

Human-centered countermeasures are equally emphasized in the literature. Employee training programs are essential for raising awareness of unauthorized AI usage and reinforcing adherence to cybersecurity policies, and AI literacy programs tailored to specific sectors, particularly education, are needed to refine assessment methods and ensure academic integrity (Balogun et al., 2025, p. 6). Digital organizational culture plays a positive moderating role in the relationship between human AI skills and organizational innovation (An, Lin, and Luo, 2024, p. 1), suggesting that organizations which cultivate a supportive digital culture can channel AI adoption toward sanctioned and innovative outcomes rather than covert workarounds. Nonetheless, challenges such as resistance to change and ethical concerns remain significant barriers to cultural alignment, even where AI adoption promises clear organizational benefits (Murire, 2024, p. 1). Collectively, the literature advocates for a governance posture that neither prohibits AI use outright nor ignores its risks, but instead creates structured pathways, through frameworks, training, technical safeguards, and cultural enablement, through which employees can adopt AI responsibly and organizations can maintain meaningful oversight.

5 Conceptual Development: Extending Shadow IT to Shadow AI

REPHRASE

Building on the previously established theoretical foundations and the insights of the SLR results, this section focuses on the conceptual linkage and analytical extension of SIT to SAI. While SIT has been extensively studied over the past decades, resulting in a well-established understanding of its underlying motivations, organizational consequences, and governance approaches, the emergence of SAI presents a relatively novel phenomenon. Existing literature on SAI is still limited in scope and fragmentation, offering only initial insights withoout a comprehensive conceptual framework.

To address this gap, this section adopts a transfer-oriented perspective. It systematically draws on the established body of knowledge on SIT and examines how its core concepts, theories, and findings can be applied to the context of SAI. This approach is based on the assumption that SAI, while technologically more advanced, represents an evolution rather than a completely independent phenomenon. As such, similar organizational drivers and bevavioral patterns are expected to persist, even though they may manifest differently due to the specific characteristics of AI technologies.

At the same time, the transfer is not assumed to be one-to-one. AI systems introduce new properties, such as increased autonomy, opacity, and scalability, that may alter both the nature and the impact of associated risks and opportunities. Therefore, the analysis not only seeks to identify parallels between SIT and SAI but also to highlight important deviations that emerge in the AI context.

By structuring the conceptual development along the dimensions of motivations, consequences, and governance approaches, this section provides a theoretically grounded foundation for the subsequent analysis. It enables a systematic exploration of SAI despite the current limitations of dedicated literatuer, while ensuring that the insights remain anchored in established research.

REPHRASE

5.1 Transferable Motivations

The motivational architecture that drives SIT adoption does not dissolve when employees transition to AI-based tools, it transfers, structurally intact, into the SAI domain. The most fundamental driver in both phenomena is the existence of an unmet need arising from a gap between what the official IT organization provides and what employees require to perform their work (Rakovic et al., 2020, p. 148; Ross, Hibbert, and Moss, 2025, p. 6). In the SIT context, this gap is rooted in the failure to provide adequate applications and the inability to fulfill business needs through legitimate channels (Györy et al., 2012, p. 6); in the SAI context, the same structural logic produces identical behavior, as

employees adopt unsanctioned AI tools in direct response to perceived inefficiencies in officially sanctioned digital infrastructures (Ross, Hibbert, and Moss, 2025, p. 2).

The productivity and performance orientation is equally transferable. SIT users are motivated by the promise of faster task completion and enhanced individual performance (Pinto, Beerepoot, and A. C. G. Maçada, 2023, p. 239), while SAI adopters are similarly driven by pressing demands for productivity gains, task automation, and augmented decision-making (Ross, Hibbert, and Moss, 2025, pp. 2, 3). Both populations perceive their unofficial tools as more useful than sanctioned alternatives (G. Mallmann and A. C. Maçada, 2016, p. 3; Chin et al., 2025, p. 2). Bureaucratic barriers and IT department sluggishness are likewise a direct motivational antecedent in both contexts: SIT emerges from formal development procedures, approval bottlenecks, and prohibitively high transaction costs of dealing with central IT (Stephan Zimmermann, Christopher Rentrop, and Felden, 2014, p. 7; Chua, Storey, and Chen, 2014, p. 9), while SAI users feel that the agility gained through AI experimentation justifies bypassing slower official approval processes (Raizada, Shakya, and Rahul, 2026, p. 1136).

Policy unawareness and non-malicious intent also transfer directly. SIT users frequently persist in non-compliant behavior because they do not understand applicable policies and do not perceive their behavior as harmful (M. Silic and A. Back, 2014, pp. 277–279); SAI adopters equally conceal their usage when formal guidance is lacking and do not act with harmful intent (Varshini and Sampathkumari, 2025, p. 2). The consumerization dynamic transfers as well: just as the proliferation of consumer-grade IT tools lowered the adoption threshold for SIT (Thatte, Grainger, and McKay, 2012, p. 8), the widespread availability of intuitive, consumer-facing AI services, accessible without expert technical knowledge (Gabriela Labres Mallmann and Antonio Carlos Gastaud Maçada, 2021, p. 436), has replicated that lowering of barriers specifically for AI tools (Puthal, Ray, and Mohanty, 2025, p. 973).

5.2 Novel Motivations

While the motivational foundations of SAI are substantially inherited from SIT, several drivers emerge that are genuinely novel and specific to the AI context. The most distinctive is the skill gap compensation dynamic: employees resort to consumer-grade AI solutions to actively compensate for technical weaknesses within their teams and use AI capabilities to offset individual skill deficits that would not be addressable through conventional software tools (Ross, Hibbert, and Moss, 2025, pp. 1–2; An, Lin, and Luo, 2024, p. 3). This represents a qualitative departure from SIT motivation, where employees sought tools that better matched their existing workflows, not tools that extended their cognitive or technical capabilities beyond their baseline competencies.

A second novel motivational dimension is the competitive and survival imperative associated with AI adoption. In the era of the digital economy, rapid technological innovation and globalization make it effectively inevitable for organizations and individuals to adopt AI in order to maintain long-term competitiveness (An, Lin, and Luo, 2024,

p. 1). This positions SAI not merely as a workaround for operational inconvenience but as a perceived strategic necessity, a framing absent from the SIT literature. SAI tools also promise competitive advantage in a way that generic shadow software does not (Puthal, Ray, and Mohanty, 2025, p. 973), and their applicability spans a qualitatively broader range of knowledge work tasks, including decision-making, communication, information processing, coordination, and value creation (Alter, 2024, p. 11), than the task-specific tools that characterize most SIT adoption.

The governance drift zone constitutes a third novel motivational enabler. Unlike SIT, which operates in contexts where formal IT policies are broadly established, SAI frequently emerges in organizational environments where AI governance frameworks simply do not yet exist or exist only in name, creating a vacuum in which policy-violating behavior cannot technically occur (Mario Silic, D. Silic, and Kind-Trüller, 2025, p. 1; Raizada, Shakya, and Rahul, 2026, p. 1138). The lack of policy awareness and vague approval mechanisms specific to AI facilitate risk-taking behavior that would face clearer constraints in a mature IT governance environment (Raizada, Shakya, and Rahul, 2026, p. 1140). Finally, the accessibility enabled by AI model repositories such as Hugging Face Hub allows non-experts to rapidly prototype and deploy sophisticated models (Puthal, Ray, and Mohanty, 2025, p. 973), enabling a depth of unauthorized technical deployment that goes well beyond what was achievable in the consumer IT consumerization wave that drove SIT.

5.3 Behavioral Shift in Knowledge Work

The transition from SIT to SAI represents more than an incremental extension of unauthorized tool adoption, it reflects a qualitative shift in how employees relate to technology within knowledge work. SIT was primarily adopted to enhance the efficiency of existing task structures: employees substituted better tools for inadequate ones, automating administrative tasks and improving access to information (White, 2023, p. 114; Koch et al., 2014, p. 2). AI tools, by contrast, are adopted to fundamentally augment cognitive work: they automate not merely repetitive manual tasks but analytical, communicative, and creative processes, drafting, summarizing, brainstorming, and decision support (Ross, Hibbert, and Moss, 2025, p. 4; Mario Silic, D. Silic, and Kind-Trüller, 2025, p. 7). SAI has been credited with sparking creativity and enabling employees to approach problem-solving in entirely new ways (Mario Silic, D. Silic, and Kind-Trüller, 2025, p. 9), a consequence that extends well beyond the efficiency gains associated with SIT.

Behaviorally, the adoption of SAI is characterized by a culture of digital improvisation in which employees feel compelled to discover and deploy their own solutions, and this culture is fueled by desires for innovation, resistance to bureaucracy, and the informal social networks through which AI tools are shared (Ross, Hibbert, and Moss, 2025, pp. 1, 3). Psychological safety and innovation-seeking are identified as the primary behavioral drivers of experimentation with unmonitored AI tools, a motivational

profile that is more cognitively and culturally grounded than the operational frustration that typically underlies SIT adoption (Raizada, Shakya, and Rahul, 2026, p. 1138). The perceived productivity gains of AI tools also outweigh compliance concerns for the average employee, particularly under task pressure (Raizada, Shakya, and Rahul, 2026, pp. 1136, 1140), creating a rationalization dynamic structurally similar to, but more psychologically entrenched than, the performance expectancy justifications documented for SIT (Nguyen, 2024, p. 505). This behavioral pattern, in which innovation effectively bypasses organizational control and top-down prohibitions are rendered largely ineffective, signals that the governance challenge posed by SAI is not merely an intensification of the SIT problem but a structurally distinct behavioral phenomenon rooted in the nature of AI itself (Raizada, Shakya, and Rahul, 2026, p. 1136).

5.4 Integrated Conceptual Framework

The findings from both systematic literature reviews enable the construction of an integrated conceptual framework that maps the evolution from SIT to SAI and identifies the structural continuities and discontinuities between the two phenomena. At the broadest level, both phenomena share a common causal architecture: they are jointly produced by a triad of enablers, motivators, and missing barriers (Klotz, Kopper, et al., 2019, p. 22). The enabling conditions, consumer-grade tool accessibility, organizational resource constraints, and low technical barriers to adoption, are present in both contexts and represent the structural preconditions without which neither SIT nor SAI would emerge at scale. The motivators, unmet needs, performance pressure, bureaucratic friction, autonomy, and policy unawareness, operate across both phenomena with the additions of AI-specific drivers (skill gap compensation, competitive necessity, governance vacuum) that elevate the intensity and breadth of SAI adoption. The missing barriers, inadequate governance, weak social norms, absent or unenforced policies, are amplified in the SAI context by the governance drift zone that characterizes the current organizational response to AI.

The consequence structures are parallel in form but divergent in severity. Both phenomena produce a dual profile of risks and benefits: cybersecurity exposure, governance disruption, and operational inefficiency on the risk side; productivity gains, innovation, and enhanced communication on the benefit side (Gabriela Labres Mallmann, Antônio Carlos Gastaud Maçada, and Andreas Eckhardt, 2018, p. 13; Ross, Hibbert, and Moss, 2025, p. 1). However, SAI introduces consequence categories with no equivalent in the SIT literature, including adversarial model attacks, algorithmic bias, sector- specific ethical liability, and knowledge leakage through generative outputs, that represent a qualitative escalation of organizational risk (Puthal, Ray, and Mohanty, 2025, p. 973; Balogun et al., 2025, p. 3; Chin et al., 2025, p. 1).

At the governance level, the framework reveals an evolutionary trajectory: the countermeasure portfolio established for SIT, spanning deterrence, formalization, technical detection, training, and user-centric need fulfillment (Rakovic et al., 2020, pp. 150–151;

Györy et al., 2012, pp. 8–9), provides a foundational template that must be substantially extended for SAI. New governance instruments, Zero Trust Architecture, AI-aware DLP, XAI, the BYOAI-GOVT framework, and supranational regulatory compliance with the EU AI Act, are required not as replacements for the SIT governance toolkit but as extensions of it, designed to address the novel risk dimensions that AI introduces (Rakovic et al., 2020, p. 144; Anthuvan et al., 2025, p. 9623; Balogun et al., 2025, p. 7). The integrated framework thus positions SAI as a second-generation manifestation of the SIT phenomenon, structurally continuous, but consequentially escalated and governance-inadequate by the standards of existing organizational practice.

6 Comparative Analysis of Shadow IT and Shadow AI

6.1 Differences in Motivational Drivers

While SIT and SAI share a common motivational bedrock, a comparative analysis reveals meaningful differences in the depth, breadth, and psychological character of their respective driver profiles. In the SIT context, motivations are predominantly operational in nature: employees adopt unauthorized tools primarily because official systems are inadequate, IT departments are too slow, or the transaction costs of formal procurement are prohibitive (Stephan Zimmermann, Christopher Rentrop, and Felden, 2014, p. 7; Keskin et al., 2021, p. 3). The motivational logic is essentially substitutive, employees replace inadequate official tools with better ones, and is anchored in concrete task-level frustrations such as system usability deficits, missing functionality, or inflexible legacy environments (Myers et al., 2017, p. 108; Nwankpa, 2024, p. 1).

SAI motivations, while sharing this substitutive logic, extend into cognitive and strategic dimensions that SIT does not reach. Employees adopt AI tools not only to substitute for missing capabilities but to extend their own cognitive capacity, compensating for skill gaps (Ross, Hibbert, and Moss, 2025, pp. 1–2; An, Lin, and Luo, 2024, p. 3), accessing creative and decision-support capabilities they could not otherwise achieve, and positioning themselves competitively within an AI-driven economy (Myers et al., 2017, p. 105). The psychological profile of SAI adoption is also distinct: psychological safety and innovation-seeking are identified as primary behavioral drivers (Raizada, Shakya, and Rahul, 2026, p. 1138), whereas SIT adoption is more typically driven by effort expectancy and performance expectancy within established task structures (Nguyen, 2024, pp. 500, 505). Furthermore, SAI benefits from a governance vacuum that SIT generally does not: vague or entirely absent AI approval mechanisms facilitate adoption in contexts where equivalent IT policy gaps would already have been closed (Raizada, Shakya, and Rahul, 2026, p. 1140; Mario Silic, D. Silic, and Kind-Trüller, 2025, p. 1). In sum, while both phenomena are rooted in the same structural misalignment between organizational IT supply and employee demand, SAI motivations are more culturally embedded, more cognitively oriented, and more strategically rationalized than those of

SIT.

6.2 Differences in Organizational Risk

SIT and SAI share a family of risks, cybersecurity exposure, data leakage, governance disruption, operational inefficiency, and regulatory liability, but differ substantially in the severity, tractability, and sectoral specificity of those risks. In the SIT domain, cybersecurity risks are primarily structural: unauthorized tools expand the attack surface, fall outside IT security oversight, and introduce malware or data sovereignty violations (Nwankpa, 2024, p. 1; M. Silic and A. Back, 2014, pp. 278–280). These risks, while serious, are broadly amenable to established IT security controls, monitoring, policy enforcement, network traffic analysis, and access restriction (Mario Silic, Barlow, and Andrea Back, 2017, p. 1029; Walterbusch, Fietz, and Teuteberg, 2017, p. 644). The operational consequences of SIT, technical debt, data inconsistencies, integration problems, and standardization failures, are similarly bounded by the scope of the unauthorized tools themselves, which typically address specific functional tasks within individual business units (Kopper, Fuerstenau, et al., 2018, p. 3).

SAI introduces a risk profile that is both broader and fundamentally harder to mitigate. At the technical level, SAI systems generate qualitatively new attack vectors with no SIT equivalent: shadow backdoor models can be engineered to facilitate adversarial attacks, and shadow data leak models can channel sensitive organizational data to unauthorized recipients in ways that conventional IT security architecture is not designed to detect (Puthal, Ray, and Mohanty, 2025, p. 973). Model hallucination, the generation of plausible but factually incorrect outputs, introduces operational risks of a kind that do not arise from conventional software tools and that can corrupt organizational decision-making in ways that remain undetected for extended periods (Raizada, Shakya, and Rahul, 2026, p. 1136). The risk of organizational knowledge leakage through generative AI is also structurally distinct from the information security risks of SIT: employees who input proprietary data into unsanctioned LLMs expose that data to third-party training pipelines and external servers in ways that conventional data loss prevention frameworks are not equipped to govern (Chin et al., 2025, p. 1; Puthal, Ray, and Mohanty, 2025, p. 2).

The ethical and sector-specific risk dimensions represent perhaps the sharpest point of differentiation. SIT governance challenges are primarily organizational and technical; SAI governance challenges extend into questions of algorithmic fairness, individual rights, and societal harm. The unchecked deployment of AI-driven credit scoring systems has produced discriminatory lending practices, while unauthorized AI diagnostic tools in healthcare risk patient confidentiality and misdiagnosis (Balogun et al., 2025, pp. 2, 3). The black-box nature of many AI systems compounds these risks by making it structurally impossible to audit how decisions were reached (Mario Silic, D. Silic, and Kind-Trüller, 2025, p. 1; Balogun et al., 2025, p. 4). No equivalent accountability gap exists in the SIT risk literature, where the consequences of unauthorized tool use, while

operationally significant, do not carry the same potential for discriminatory or legally liable outcome at scale.

6.3 Differences in Governance Approaches

The governance approaches developed for SIT provide a template that is partially transferable to SAI, but the novel risk dimensions of AI necessitate a substantially extended and redesigned governance architecture. SIT governance literature converges on a spectrum from deterrence-focused approaches, formal and informal sanctions, policy enforcement, and active monitoring, to enabling approaches that legitimize user-driven IT through formalization, co-governance, and need fulfillment (Györy et al., 2012, pp. 8–9; Stephan Zimmermann, Christopher Rentrop, and Felden, 2016, p. 9). The central governance insight from the SIT literature is that prohibition alone is ineffective: strict policies push SIT further into secrecy rather than eliminating it (Steffi Haag and Andreas Eckhardt, 2017, p. 472; Klesel et al., 2019, p. 856), and the most effective responses combine technical controls with user-centric approaches that address the underlying needs driving adoption (Gabriela L. Mallmann, Antonio Carlos Gastaud Maçada, and Oliveira, 2018, p. 26; Kopper, Klotz, et al., 2019, p. 14).

This core insight transfers directly to SAI, top-down prohibitions of AI tools have similarly been found to be largely ineffective, driving usage further underground and creating a culture where innovation bypasses control (Raizada, Shakya, and Rahul, 2026, p. 1136). However, the governance instruments required to operationalize this principle in an AI context are substantially more sophisticated. Whereas SIT governance relies on network traffic monitoring, software standardization, and access controls (M. Silic and A. Back, 2014, p. 278; Koch et al., 2014, p. 9), SAI governance demands AI-specific technical safeguards including Zero Trust Architecture, AI-aware DLP, content filtering, AI traffic monitoring, and automated anomaly detection (Puthal, Ray, and Mohanty, 2025, pp. 973, 974; Whitford, Marcel, and McCall, 2025, p. 1). XAI represents an entirely new governance instrument with no SIT counterpart, enabling organizations to make AI decision-making interpretable and to identify biases or errors produced by shadow systems (Balogun et al., 2025, p. 7).

At the framework level, SIT governance has developed structured typologies, including IT-control, user-driven, and user-oriented approaches, as well as co-governance and instance-level responsibility allocation models (Györy et al., 2012, pp. 8–9; Klotz, Kopper, et al., 2019, p. 30), that provide a conceptual vocabulary for governing SAI. The BYOAI-GOVT framework extends this vocabulary into the AI domain, integrating task-risk zoning across Enable, Regulate, and Restrict categories with a four-level organizational maturity scale (Anthuvan et al., 2025, p. 2693). SAI governance also operates under a supranational regulatory layer that has no equivalent in the SIT governance landscape: the EU Artificial Intelligence Act categorizes AI applications by risk level and imposes binding compliance obligations, while the NIST AI Risk Management Framework mandates regular compliance audits and risk assessments (Balogun

et al., 2025, p. 7). These external regulatory instruments shift the governance calculus for SAI beyond organizational risk management into the domain of legal liability and regulatory compliance, a dimension largely absent from the SIT governance literature.

Finally, the human and cultural dimensions of governance differ in emphasis. SIT governance emphasizes security policy communication, training, and the removal of structural barriers as mechanisms for closing the policy awareness gap (Silvius and Dols, 2012, p. 12; M. Silic and A. Back, 2014, p. 279). SAI governance equally requires training and awareness programs, but extends these into domain-specific AI literacy, including education-specific programs to ensure academic integrity (Balogun et al., 2025, p. 6), and the cultivation of a digital organizational culture that actively channels AI skills toward sanctioned innovation (An, Lin, and Luo, 2024, p. 1). Resistance to change and concerns about ethical alignment remain significant governance barriers specific to the AI context, suggesting that SAI governance must address not only the behavioral dynamics of non-compliance but the deeper cultural and ethical dimensions of AI adoption that extend well beyond the scope of conventional IT governance frameworks (Murire, 2024, p. 1).

7 Discussion

8 Limitations

While this review provides a comprehensive overview and valuable conceptual insights into motivations, consequences, and governance approaches of SAI as a emerging topic, it is not without limitations.

First, the literature search was limited to four databases and specific search strings, which may have resulted in the exclusion of other relevant studies that use different terminology or appear in journals not covered by the databases. Although backward and forward searching was conducted to mitigate this risk, it cannot be completely ruled out.

Second, the review was conducted by a single researcher, which may introduce potential bias in the selection, coding, and interpretation of the literature. While efforts were made to ensure rigor and objectivity, the absence of multiple reviewers may limit the reliability of the presented findings.

Third, the conceptual development and comparative analysis are based on the assumption of a transferability of insights from SIT with the risk of oversimplification in the context of AI due to the different technological characteristics and types of AI tools.

Fourth and finally, the review is limited to literature published to the current date (March 2026), which may not capture all relevant facets of the emerging field of SAI.

9 Future Research Directions

Given the findings and limitations outlined above, there is ample scope for future research on this topic. Empirical studies are needed to validate the transferability of SIT to SAI and to explore the AI-specific characteristics and their impact on organizational processes and outcomes. Qualitative studies, such as case studies or interviews, could provide deeper insights into the changes in organizational practices and employee behavior, while quantitative studies could test the relevance and strength of identified relationships.

Although many forms of SAI can be derived from and build upon the foundations of SIT, AI poses some unique risks that have either already been studied, have not yet been identified, or have yet to emerge. For this reason, more in-depth and specific long-term research is needed.

Overall the field of SAI is still in its early stages, calling for a wide range of research areas (e.g. socio-technical, cybersecurity, ethics) to be explored.

10 Conclusion

In der Schlussfolgerung sollen

• die Themenstellung • der gewählte Ansatz • die Ergebnisse der Arbeit • eine kritische Stellungnahme/Einschätzung • nächste Schritte

deutlich werden. Hinweis: Die Schlussfolgerung sollte mit der Zusammenfassung bzw. dem Abstract und der Einleitung abgeglichen werden. Es sollte immer eine Zusammenfassung der wesentlichen Erkenntnisse der eigenen Arbeit sein, die den Forschungsbeitrag darstellt. Der Umfang der Schlussfolgerung sollte ähnlich wie die Einleitung ca. 5 % der gesamten Arbeit betragen. (= ca. 2 Seiten)

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A Appendix A